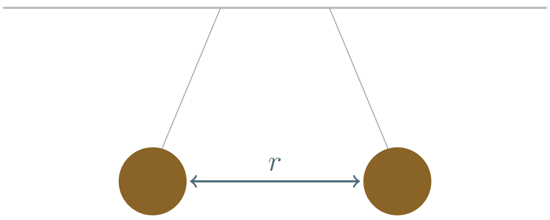
How Does Repulsion Fade?

Name: ____________________

Two charged balloons, an imperfect apparatus, and a search for the shape of a law rather than confirmation of one supplied in advance.



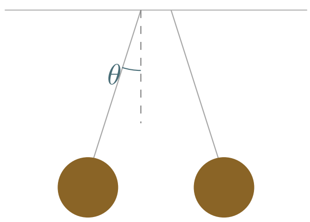
The question. Two similarly charged balloons repel one another. Construct the strongest model you can for how the force depends on separation.

Working rule: Do not assume the exponent in advance. Progress includes a defensible measurement model, a comparison between candidate laws, an uncertainty, a reason for rejecting data, a model limitation, or a sharper next question.

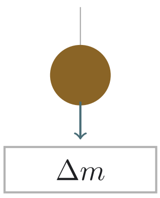
1. Two ways to see the interaction

force inferred from equilibrium

The same electric interaction can be measured in two different ways.



horizontal deflection



vertical balance

For the suspended balloon, resolve the thread tension into horizontal and vertical components:

$$T \sin\theta = F, \qquad T\cos\theta = mg.$$

Show that

$$F = mg\tan\theta.$$

Working:

For the vertical balance method, explain why a change in displayed mass corresponds to

$$F = \Delta m\, g.$$

Which method gives the more direct force reading? Which makes the force geometry more visible?

2. Define what you are measuring — the separation problem

A balloon is not a mathematical point. What should count as the separation r?

Definition of r	Advantage	Possible weakness
centre to centre		
surface to surface		
nearest charged regions		
other:		

Choose one operational definition for your investigation and justify it.

Record the balloon radius R and the range of separations you intend to use.

$R =$ ____________ $r_{\text{min}} =$ ____________ $r_{\text{max}} =$ ____________

Why might the dimensionless ratio r/R be more informative than r alone?

3. Gather first measurements

look for a pattern before fitting

Keep the charges as nearly unchanged as practical. Measure force at several separations. Include repeats where possible.

r	F_1	F_2	F_3	mean F

Describe the pattern without naming a law.

When the separation approximately doubles, the force becomes approximately:

one half | one quarter | one eighth | other

Evidence:

4. Compare candidate power laws

earn the exponent

Use the general model

$$F(r) = \frac{A}{r^n},$$

where A absorbs the charge strengths and other quantities held fixed.

Doubling test. Show that

$$\frac{F(r)}{F(2r)} = 2^n.$$

Working:

Use one pair of your measurements to estimate n:

$$n = \frac{\log[F(r_1)/F(r_2)]}{\log(r_2/r_1)}.$$

$r_1 =$ __________, $r_2 =$ __________, $F_1 =$ __________, $F_2 =$ __________, $n \approx$ __________

Constant-product test. Calculate rF, r^2F, and r^3F. The most nearly constant column supports the corresponding exponent.

r	F	rF	r^2F	r^3F

Which column is most nearly constant? State how you judged "most nearly".

5. Linearise only after the simpler tests — log-log reasoning

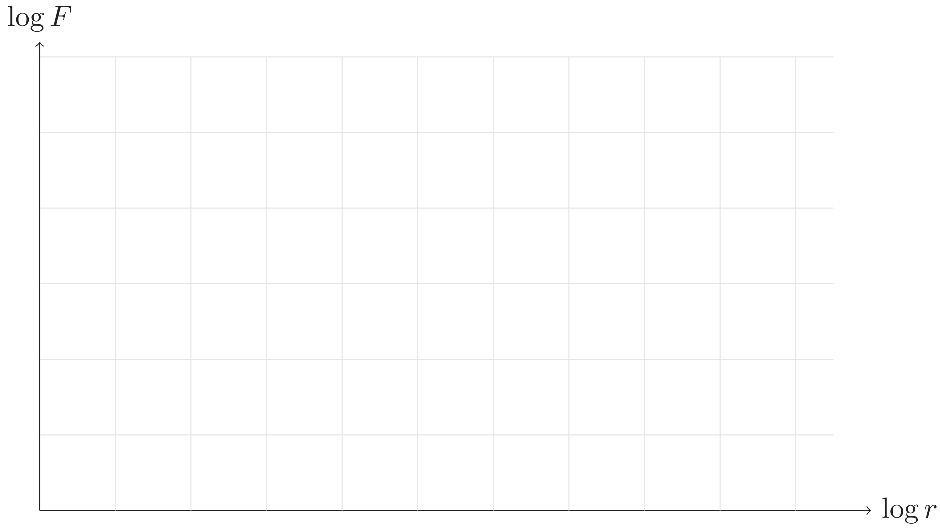
Taking logarithms gives

$$\log F = \log A - n \log r.$$

On a graph of $\log F$ against $\log r$:

a) the gradient should be ________;

b) the intercept should be ________.



Estimated gradient = ________, so $n \approx$ ________.

Does the fitted exponent change if you omit the closest-separation points? Record what happens and why that matters.

6. The crucial complication — a balloon is not a point charge

For each feature, predict how it might alter the measured relationship.

Large compared with r

Deformable

Electrically polarisable

Unevenly charged

Charge leakage

Humidity

Air currents

Charge redistribution as balloons approach

A poor inverse-square fit does not by itself refute electrostatics. Explain two other possible interpretations.

1. ______________________________

2. ______________________________

Propose an operational criterion for when the point-charge approximation becomes defensible, using r/R.

7. Distinguish error from model failure

design a discriminating test

Choose one suspected source of error or model limitation. Design a test that would change one factor while leaving others as constant as practical.

Suspected cause

Variable deliberately changed

Variables controlled

Prediction if this cause matters

Observation that would count against it

Would repeating the experiment with smaller charged spheres strengthen the point-charge model? Explain.

8. From force to potential

one power changes

The radial force and potential energy are related by

$$F_r = -\frac{dU}{dr}.$$

For a repulsive force magnitude

$$F(r) = \frac{k}{r^n},$$

show that, for $n \neq 1$ and with $U(\infty) = 0$,

$$U(r) = \frac{k}{(n-1)r^{n-1}}.$$

Working:

Complete the table.

Force law	Potential-energy law	Difference in exponent
$F \propto 1/r$		
$F \propto 1/r^2$		
$F \propto 1/r^3$		

Why does an inverse-square force correspond to an inverse-distance potential?

9. Why might the exponent be two? geometry enters after measurement

Influence from a point source spreads through spherical surfaces whose area is

$$A(r) = 4\pi r^2.$$

If the same total flux crosses every sphere, explain why flux per unit area scales as $1/r^2$.

List the assumptions hidden inside this argument.

1. ________________
2. ________________
3. ________________
4. ________________

Why does this geometric argument motivate, rather than by itself prove, Coulomb's law?

10. Productive stuck — record what the investigation has genuinely established

Complete at least five rows.

Observation	As r increased, the apparatus showed...
Measurement model	We inferred force from...
Candidate law	The data most nearly support...
Evidence	The strongest calculation or graph was...
Model limitation	This conclusion depends on...
Range dependence	The fitted exponent changed/did not change when...
Point-charge criterion	The approximation becomes more credible when...
Potential interpretation	If $F \propto r^{-n}$, then $U \propto$...
Next question	The next measurement that would discriminate between models is...

11. Final model statement — separate measurement, inference, and assumption

Write one compact paragraph containing all four elements:

1. what was directly measured;
2. how force and separation were operationally defined;
3. which power law was best supported over which range;
4. why the conclusion should not be overgeneralised to all charged bodies.

Success criterion: A strong investigation does not merely report an exponent. It shows how the exponent was extracted, where it is credible, what idealisations make it meaningful, and what evidence could still overturn the model.

12. Core results

what this enquiry actually earned

The relations established along the way.

$$F(r) = \frac{A}{r^n} \qquad \frac{F(r)}{F(2r)} = 2^n \qquad U(r) = \frac{k}{(n-1)r^{n-1}} \qquad A(r) = 4\pi r^2$$

n was never assumed; it was extracted from measurements, and the extraction method is as much a result here as the number itself.

Concepts earned, in order.

- Separation, r, is an operational choice, not a given, once the objects have real size.
- The doubling test, the constant-product test, and the log-log gradient are the same test at three levels of formality.
- A poor fit near small r is evidence about where the point-charge idealisation breaks down, not evidence against electrostatics.
- Potential energy falls off one power more gently than force; the two exponents differ by exactly one.
- A geometric flux argument motivates an inverse-square law without, on its own, proving Coulomb's law.